

Read the following directions carefully. DO NOT turn to the next page until the exam has started.

Write your name and section number at the top right of this page:

Class	Section Number
Miranda 8:50	001
Miranda 9:55	002
Sahifa 1:20	003
Sahifa 2:25	004

As you complete the exam, write your initials at the top right of each other page.

When the exam start time is called, you may turn the page and begin your exam.

If you need more room, there is a blank page at the end of the exam, or we can give you some scratch paper.

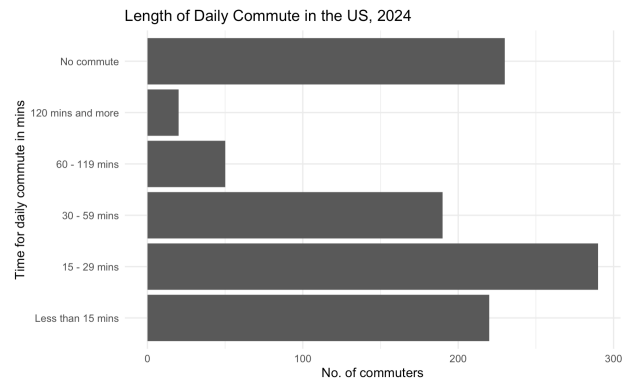
Some multiple choice questions are "Select ONE" while others are "Select ALL that apply". Pay attention to the question type and only mark one option if it says "Select ONE". Fill in the circles completely. Only highlight the circles of the selected options and leave the others completely blank.

If you finish early, you can hand your exam to your instructor or TA and leave early.

Otherwise, stop writing and hand your exam to your instructor or TA when the exam stop time is called.

1. The dataframe `commute_time` (with 1000 rows and 1 column) records the time taken to commute to work as a categorical variable `time_taken`. The code below (some details omitted) creates a graph to count the number of people in each category.

```
commute_time %>%
  group_by(time_taken) %>%
  summarize(count = n()) %>%
  ggplot(aes(x = time_taken, y = count)) +
  geom_col()
```



- (a) (4 points) You want to color the inside of the bars according to the commute time category and outline all of the bars in black. Which aesthetics do you need to add to your plot? **Select All that apply.**

☐ `color` as a variable aesthetic
 ☒ `fill` as a variable aesthetic
 ☒ `color` as a constant aesthetic
 ☐ `fill` as a constant aesthetic

- (b) (2 points) Identify a different geom that could produce the same plot without needing to tidy the data or manually create this new variable. **Select ONE.**

☐ `geom_point()`
☐ `geom_histogram()`
☒ `geom_bar()`
☐ `geom_line()`

2. (5 points) The Language Fluency dataset, saved in the environment as `Language`, contains student names, each parent's first language, and indicator variables for five languages. Each language column specifies whether the individual speaks that language (e.g., TRUE/FALSE).

```
A tibble: 50 × 8
  Name      Mom_speaks Dad_speaks English Chinese French Arabic Spanish
<chr>      <chr>      <chr>      <lgl>  <lgl>  <lgl>  <lgl>  <lgl>
Henry Todd English    English    TRUE   FALSE  FALSE  FALSE  TRUE
Ren Xiao   Japanese   Chinese    TRUE   TRUE   FALSE  FALSE  FALSE
Wendy Li   Chinese    English    TRUE   FALSE  FALSE  FALSE  TRUE
Taylor Leight English    English    TRUE   FALSE  TRUE   FALSE  TRUE
Joseph Lopez Spanish    English    TRUE   FALSE  FALSE  FALSE  FALSE
Aaron Sato English    Japanese   TRUE   TRUE   FALSE  FALSE  FALSE
.... plus 44 more rows
```

The *Undergraduate Major and GPA* dataset saved in the environment as `Major` contains the names of 20 students (15 of them are also present in the `Language` dataset), their Undergraduate majors, and other columns.

```
A tibble: 20 × 5
  Name      Undergrad_major      Year_of_graduation  GPA Grad_School
<chr>      <chr>                  <dbl>  <dbl> <chr>
Henry Todd Computer Science      2020  3.45 YES
Monica Keale Biology                2021  2.78 NO
Wendy Li   Psychology                2027  3.1  NA
Taylor Leight Business Administration 2022  3.92 YES
Ezequial McPherson English Literature 2024  3.88 NO
Joseph Lopez Mathematics          2026  3.98 YES
.... plus 14 more rows
```

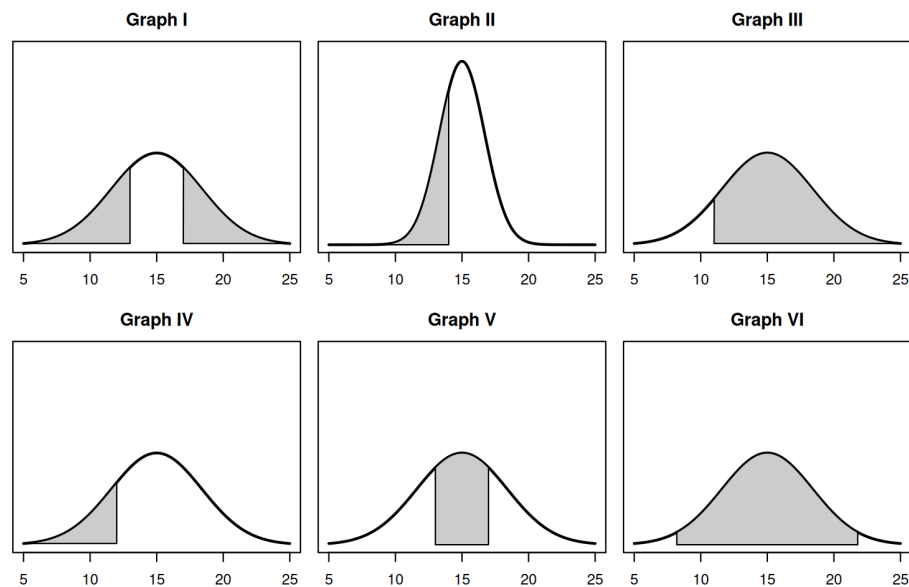
Working with the `Language` and `Major` and predict the dimension of the output of the following joins. Assume that `join_by` is automatically set to join correctly by the student's name.

Join function	Rows	Columns
<code>semi_join(Major, Language)</code>	<u>15</u>	<u>5</u>
<code>full_join(Major, Language)</code>	<u>55</u>	<u>12</u>
<code>left_join(Major, Language)</code>	<u>20</u>	<u>12</u>
<code>left_join(Language, Major)</code>	<u>50</u>	<u>12</u>
<code>anti_join(Language, Major)</code>	<u>35</u>	<u>8</u>

3. (5 points) The amount of peanuts in a random candy bar from a production line is normally distributed with mean 15 grams and standard deviation 3.4 grams.

You are given several graphs (labeled I-VI) that have the same axes, several qualitative descriptions (labeled A-F) and a table with R code. Complete the table by matching the R code to the correct description and the correct shaded area in the graph. One of the graphs and one of the descriptions will not be used in the table, and all of the other graphs/descriptions will be used only once.

Graphs:



Descriptions:

- A: A randomly chosen candy bar has over 11 grams of peanuts.
 B: The amount of peanuts in a random candy bar is within two standard deviations of the average.
 C: The amount of peanuts in a random candy bar is not within two grams of the average.
 D: The average amount of peanuts in 2 randomly chosen candy bars is less than 14 grams.
 E: The average amount of peanuts in 4 randomly chosen candy bars is less than 14 grams.
 F: A randomly chosen candy bar has less than 12 grams of peanuts.

R code	Description (A-F)	Graph (I-VI)
<code>pnorm(12, 15, 3.4)</code>	<u> F </u>	<u> IV </u>
<code>1 - pnorm(11, 15, 3.4)</code>	<u> A </u>	<u> III </u>
<code>2*pnorm(13, 15, 3.4)</code>	<u> C </u>	<u> I </u>
<code>1 - 2*pnorm(8.2, 15, 3.4)</code>	<u> B </u>	<u> VI </u>
<code>pnorm(14, 15, 3.4/2)</code>	<u> E </u>	<u> II </u>

4. (4 points) Identify whether the following statements about hypothesis testing are TRUE or FALSE.

(a) A p-value of 0.05 means that there is a 95% chance that the alternative hypothesis is true.

☐ TRUE

☒ FALSE

(b) When conducting a hypothesis test, start by assuming the null hypothesis H_0 is true.

☒ TRUE

☐ FALSE

(c) The p-value is the probability of observing a test statistic at least as "extreme" as the observed statistic if H_0 were true.

☒ TRUE

☐ FALSE

(d) In hypothesis testing, the null and alternative hypotheses are statements about the value of the observed sample.

☐ TRUE

☒ FALSE

(e) It is best practice to choose α for a hypothesis test before looking at the data.

☒ TRUE

☐ FALSE

(f) If the test statistic is less than α , we have a significant result and we decide to reject H_0 .

☐ TRUE

☒ FALSE

(g) α is the significance level and also the probability of falsely rejecting H_0 .

☒ TRUE

☐ FALSE

(h) Based on the p-value, we either accept the null or the alternative hypothesis.

☐ TRUE

☒ FALSE

5. For each of the following scenarios, select which type of hypothesis test or statistical method would be most appropriate.

Select ONE each.

(a) (2 points) 100 employees at a company have their performance recorded before and after completing a training program. The company wants to compare the average employee performance before and after the program.

☐ Equal variances T test

☐ Welch T test

☒ Paired T test

☐ Linear regression

(b) (2 points) An ice cream shop lets customers choose between either a cup or a cone. They want to know which option is more popular, so they take a sample of 50 customers and record their choice.

☒ One-sample proportion test

☐ Two-sample proportion test

☐ One-sample T test

☐ Welch T test

6. The dataset 'bm_sample' records the sex, age, and completion time for several runners of the Bostom Marathon in both 2010 and 2011.

BibNum <dbl>	Year <dbl>	sex <chr>	Age <dbl>	age_range <chr>	total_time <dbl>
5668	2011	male	53	50-54	188.55
2863	2011	male	53	50-54	199.19
14490	2010	female	33	18-34	240.84
18743	2011	female	41	40-44	247.73
13823	2011	female	28	18-34	209.25
27033	2010	female	35	35-39	295.54

- (a) (6 points) We want to create the following the summary table for all the male participants:

```
A tibble:2 × 5
  Year   n_male avg_time sd_time  n_total
<dbl> <int>   <dbl>   <dbl>   <dbl>
2010   2802   220.5967  37.76457  4940
2011   2942   222.8564  39.66779  5060

bm_sample %>%
  filter(sex == male) %>%
  group_by("Year") %>%
  summarise(n_male = count(),
            avg_time = mean(total_time),
            sd_time = sd(total_time)) %>%
  mutate(n_total = c(4940, 5060))
```

The code above is supposed to produce the summary table, but it does not work correctly. Identify the problems with the code and **Select ALL** fixes from the list below.

- ☒ Change `sex == male` to `sex == "male"` in `filter()`.
- ☐ Change `sex == male` to `sex == female` in `filter()`.
- ☒ Change `"Year"` to `Year` in `group_by()`.
- ☐ Change `"Year"` to `c("Year", "sex")` in `group_by()`.
- ☒ Change `count()` to `n()` in `summarise()`.
- ☐ Change `summarise()` to `mutate()`.

- (b) (5 points) The summary table from the previous question has been saved as 'bm_summary'. Complete the table below to show the output of the following `pivot_longer()` operation. The gray shaded boxes are for the column names.

```
bm_summary %>%
  select(c(Year, n_male, n_total)) %>%
  pivot_longer(cols = n_male:n_total)
```

	Year	name	value
1	2010	n_male	2802
2	2010	n_total	4940
3	2011	n_male	2942
4	2011	n_total	5060

7. Continue working with the summarized Boston Marathon data from the previous problem. We want to test the hypothesis, "There were more male participants in the 2010 Boston marathon than the 2011 Boston marathon".

Let p_1 be the proportion male participants from 2010 Boston marathon and p_2 be the proportion of male participants from 2011 Boston marathon. The sample of Boston Marathon dataset had 4940 rows for 2010 and 5060 rows for 2011.

- (a) (2 points) What are the null and alternative hypotheses for this test? **Select ONE.**

- ☐ $H_0 : p_1 - p_2 = 0$ versus $H_A : p_1 - p_2 \neq 0$
☐ $H_0 : \hat{p}_1 - \hat{p}_2 = 0$ versus $H_A : \hat{p}_1 - \hat{p}_2 < 0$
☒ $H_0 : p_1 - p_2 = 0$ versus $H_A : p_1 - p_2 > 0$
☐ $H_0 : \hat{p}_1 - \hat{p}_2 < 0$ versus $H_A : \hat{p}_1 - \hat{p}_2 = 0$

- (b) (5 points) Based on the summary table in the previous question, **give an expression for the test statistic** and identify which part of the test statistic corresponds to the **standard error**. Also, **identify the null distribution** for the test. You do not need to simplify your answer and can leave it as an expression.

Solution: From the data: in 2010 there were 2802 male participants out of 4940, and in 2011 there were 2942 male participants out of 5060. Since we are testing for an equality of proportions, we need to calculate a pooled proportion.

$$\hat{p} = \frac{2802 + 2942}{4940 + 5060}$$

The test statistic is

$$z = \frac{\frac{2802}{4940} - \frac{2942}{5060}}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{4940} + \frac{1}{5060}\right)}}$$

and the standard error is the term in the denominator of the test statistic. Finally, the null distribution is $N(0, 1)$.

- (c) (2 points) Let TS be the test statistic in (b). Which line of R code below can be used to calculate the p-value for the test? **Select ONE.**

- ☒ `pnorm(TS, lower.tail = F)`
☐ `pbinom(TS, 4940, 0, lower.tail = F)`
☐ `pnorm(TS)`
☐ `pbinom(TS, 5060, 0.5)`

- (d) (3 points) The p-value from part (c) comes out to be 0.9247186. What are the correct conclusions for this test if $\alpha = 0.05$? Report both a **decision** and an **interpretation in context**.

Solution: Because the p-value is larger than α , we fail to reject the null. We do not have evidence that there was a higher proportion of male Boston Marathon participants in 2010 than 2011.

8. Librarians recorded the amount of time spent on their public computers by 50 random users. A 95% T confidence interval for the average duration of computer use (in minutes) is reported as:

(62, 78)

Consider using the same data to conduct a hypothesis test for μ , the average minutes of computer use at the library.

- (a) (2 points) Which pair of hypotheses can be used to test whether the average duration is different from 72 minutes? **Select ONE.**

- ☐ $H_0 : \bar{X} = 72$ versus $H_A : \bar{X} \neq 72$
☐ $H_0 : \bar{X} \neq 72$ versus $H_A : \bar{X} = 72$
☒ $H_0 : \mu = 72$ versus $H_A : \mu \neq 72$
☐ $H_0 : \mu \neq 72$ versus $H_A : \mu = 72$

- (b) (4 points) Based on the confidence interval given above, write a numeric expression for the **test statistic** for the test in (a). Assume that the critical value used to construct the interval is 2. You do not need to simplify your answer.

Solution: Using the confidence interval formula $\text{pt est} \pm \text{cv} \times \text{se}$, we know that the point estimate is 70, the midpoint of the interval. The margin of error is 8, and so $\text{cv} \times \text{se} = 8 \rightarrow \text{se} = 4$. The test statistic is

$$t_{\text{obs}} = \frac{70 - 72}{4}$$

- (c) (2 points) Let TS be the test statistic in (b). Which line of R code below can be used to calculate the p-value for the test? **Select ONE.**

- ☐ `pt(TS, df = 48)`
☐ `pt(TS, df = 48, lower.tail = F)`
☐ `2*pt(TS, df = 48)`
☐ `2*pt(TS, df = 48, lower.tail = F)`
☐ `pt(TS, df = 49)`
☐ `pt(TS, df = 49, lower.tail = F)`
☒ `2*pt(TS, df = 49)`
☐ `2*pt(TS, df = 49, lower.tail = F)`

- (d) (3 points) Without knowing the p-value, what conclusions can we draw at the 5% level about the average duration being 72? Briefly explain your answer.

Solution: Because the confidence interval covers the value 72, we know that we must fail to reject the null at the 5% level. We do not have evidence that the average is different from 72.

9. Researchers wish to compare the efficiency of two vehicle models. They take a random sample of each model and record the miles per gallon (MPG) of each vehicle. Summaries of the data are given in the table below.

	Mean	Standard deviation	Sample size
Group 1	32.1	4.82	32
Group 2	29.7	3.96	32

Because the two sample SDs are similar, the researchers will assume that the two groups have equal variance.

- (a) (4 points) Other than the equal-variance assumption, what other assumptions must the researchers make for a two-sample T test to be valid? **Select ALL that apply.**
- ☐ Both populations are approximately normal.
 - ☒ The sample means of both populations are approximately normal.
 - ☒ The two groups are independent of each other.
 - ☐ The two groups have the same center.
- (b) (3 points) Write a **numeric expression for s_p , the estimated pooled standard deviation** for both groups. You do not need to simplify your answer.

Solution:

$$s_p = \sqrt{\frac{4.82^2(32 - 1) + 3.96^2(32 - 1)}{(32 + 32 - 2)}}$$

- (c) (2 points) Now, suppose the researchers do not want to make the equal variances assumption. What is true about w , the Welch degrees of freedom? **Select ONE.**
- ☒ w is less than $32 + 32 - 2$.
 - ☐ w is equal to $32 + 32 - 2$.
 - ☐ w is greater than $32 + 32 - 2$.
- (d) (3 points) Would it be valid to compare the two groups above using a paired T test? Briefly explain why or why not.

Solution: No, because we are comparing two separate vehicle models, we should treat the groups as being independent. There is no meaningful connection between the pairs of vehicles in each group.

10. Researchers surveyed 25 individuals about the number of hours of TV they watch per week (X) and their overall life satisfaction out of 100 (Y). A partial `summary()` output for a linear model for life satisfaction based on hours of TV watched is given below.

Residuals:

Min	1Q	Median	3Q	Max
-21.444	-10.520	2.788	6.068	34.759

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	60.7044	6.0025	10.113	6.18e-10 ***
x	-0.9952	0.9398	-1.059	0.301

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- (a) (3 points) Consider performing a test to identify a positive relationship between hours of TV watched and life satisfaction. **Write a null and alternative hypothesis** for this test, and **identify the null distribution**.

Solution: The hypotheses to test for a positive linear relationship are

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 > 0$$

and the null distribution is a T with $n - 2 = 23$ degrees of freedom.

- (b) (2 points) The p-value for the test in (a) is approximately equal to: **Select ONE**.

☐ 0.3 / 2
☐ 0.3

☐ 1 - 0.3
☒ 1 - (0.3 / 2)

- (c) (4 points) The estimated slope $\hat{\beta}_1$ is negative. Assuming the linear model is correct, which of the following statements are correct conclusions from this fact? **Select ALL that apply**.

- ☒ The correlation between hours of TV watched and life satisfaction is negative.
☒ Individuals in the data who watched more TV than average tend to have less life satisfaction than average.
☐ Every individual in the data who watched more TV than average must have had less life satisfaction than average.
☐ Watching more TV causes people to have less life satisfaction.

- (d) (2 points) Consider the residuals of the fitted linear model. The data point with the residual of the largest magnitude lies: **Select ONE**.

- ☐ Below the fitted line.
☐ On the fitted line.
☒ Above the fitted line.

11. A regression line is fit on 45 (x, y) pairs. Summaries of the data and the regression line are given below.

	Sample Mean	Sample Standard deviation
X	-5.8	7.48
Y	-1.9	4.56

$$\hat{y} = 0.75 + 0.46(x)$$

- (a) (3 points) Write a numerical expression for r , the correlation between X and Y . You do not need to simplify your answer.

Solution: We have:

$$\hat{\beta}_1 = r \left(\frac{s_y}{s_x} \right) \quad (1)$$

$$0.46 = r \left(\frac{4.56}{7.48} \right) \quad (2)$$

$$r = 0.46 \left(\frac{7.48}{4.56} \right) \quad (3)$$

Consider using the linear model to predict a value of y given x . Four interval estimates are made:

- A: a 95% prediction interval for $y \mid x^* = -6$
- B: a 95% confidence interval for $y \mid x^* = -8$
- C: a 95% confidence interval for $y \mid x^* = -6$
- D: a 90% confidence interval for $y \mid x^* = -6$

- (b) (2 points) What is the relationship between the point estimate of intervals A and B? **Select ONE.**

- ☐ The point estimate for A is smaller than the point estimate for B.
☐ The point estimate for A is the same as the point estimate for B.
☒ The point estimate for A is larger than the point estimate for B.

- (c) (2 points) What is the relationship between the point estimate of intervals C and D? **Select ONE.**

- ☐ The point estimate for C is smaller than the point estimate for D.
☒ The point estimate for C is the same as the point estimate for D.
☐ The point estimate for C is larger than the point estimate for D.

- (d) (2 points) What is the relationship between the critical value of intervals C and D? **Select ONE.**

- ☐ The critical value for C is smaller than the critical value for D.
☐ The critical value for C is the same as the critical value for D.
☒ The critical value for C is larger than the critical value for D.